Rishav Singh

+91 83200 61928 | rishav4805@gmail.com | rishav408 | rishavsingh408 | rishavsingh4805.web.app

Summary

B.Tech AI & Data Science student at MIT-WPU who enjoys building things that actually work — from RAG pipelines and local LLM systems to computer vision APIs. Comfortable taking a project from idea to deployed product using Python, LangChain, and TensorFlow. Looking for ML/AI or Data Science roles where I can keep learning fast and ship real solutions.

Technical Skills

Core Domains: Machine Learning, Generative AI, NLP, Computer Vision, RAG, LLM-based Systems

Languages: Python, SQL, JavaScript, HTML/CSS

ML & AI Frameworks: scikit-learn, TensorFlow/Keras, PyTorch, LangChain, Ollama

LLM & GenAI: Prompt Engineering, Embeddings, LLM API Integration (Gemini, OpenAI), Context Management

Data & CV: NumPy, pandas, OpenCV, Matplotlib, NLTK, spaCy

Backend & APIs: Flask, Streamlit, REST API Development, Model Inference Pipelines

Databases: SQLite, Vector Databases (FAISS, Chroma)

Tools & Deployment: Git, GitHub, Docker, Jupyter, Firebase Hosting

Projects

NyayaAI | *LLM, RAG, LangChain, Flask, Gemini API, NLP*

- Built a full-stack AI legal assistant that lets users upload documents and ask legal questions — answers are grounded in the actual document context via RAG, not just LLM guessing.
- Designed a stateless Flask API backend that ties together Gemini LLM reasoning, document retrieval, and web search fallback in a single clean pipeline.
- Enforced a retrieval-first strategy that significantly reduced hallucinated responses; supports multi-document uploads with dynamic context injection per query.

ResolvAI | *Local LLMs, LangChain, Ollama, Streamlit, SQLite*

- Built a fully offline AI complaint resolution tool for small businesses — no API keys, no internet, no data leaving the machine; runs entirely on local Llama3/Mistral via Ollama.
- Fed business profiles and policy documents into LangChain chains so the model responds in the brand's tone, not just a generic apology template.
- Achieved sub-3s response latency on CPU-only inference with conversation history persisted across sessions via SQLite.

Tulsi Disease Detection | *CNNs, TensorFlow/Keras, OpenCV, REST API*

- Trained custom CNN models on labelled plant disease image datasets, reaching **91%+ validation accuracy** — automated the full training and model-comparison pipeline.
- Wrapped the best-performing model in a REST API for real-time inference; built a browser frontend that shows the disease prediction, confidence score, and treatment steps.

VedaTale | *LLM APIs, Flask, Firebase, HTML/CSS/JS*

- Built an AI storytelling platform where users submit a prompt and get back a full narrative — Flask backend handles story generation via LLM APIs, Firebase hosts the frontend.
- Added a multi-page gallery and story sharing features; structured backend for easy scale-up to cloud deployment.

Education

MIT World Peace University (MIT-WPU) — Pune, Maharashtra

B.Tech — Computer Science Engineering (AI & Data Science) — CGPA: 7.43 — *Aug 2023 – Present*

Certifications

- **AI Fluency for Students** — Anthropic, 2026
- **Claude Code in Action** — Anthropic, 2026
- **Artificial Intelligence Fundamentals** — IBM SkillsBuild, 2025
- **Foundation: Introduction to LangChain (Python)** — LangChain Academy
- **Critical Thinking & Problem Solving** — Rochester Institute of Technology (RITx), Feb 2024